TimeSlice F1 API (1.0.0)

Welcome to **TimeSlice**, a high-performance analytical engine for Formula 1 telemetry.

This API is designed to bridge the gap between raw track data and actionable racing insights. By choosing specific data points across historical and modern sessions, TimeSlice allows you to dive deep into lap-by-lap performance, tyre degradation trends, and strategic execution.

WARNING: All endpoint Require a Login so please sign up first. No data after 2023 is available.

Explore the endpoints below to start generating your own racing insights.

Account Management

Register a New User

Creates a new user account in the system. The password will be securely hashed before being stored in the database. Once registered, the user can use their credentials to obtain a JWT access token.

REQUEST BODY SCHEMA: application/json
required

email required	string <email> (Email) The user's unique email address
password required	string (Password) >= 8 characters Account password (minimum 8 characters)

POST /api/v1/accounts/register

Request samples

Payload

Content type
application/json

Copy

```
{
    "email": "lewis.hamilton@mercedes.com"
    "password": "SuperSecretPassword123!"
}
```

Response samples

201 422

Content type
application/json

Copy

```
{
    "email": "lewis.hamilton@mercedes.com"
    "id": 1
}
```

Delete Account

Permanently deletes the currently authenticated user account. **Warning:** This action is irreversible and will cascade to delete all associated lap summaries and curated data.

AUTHORIZATIONS: > *OAuth2PasswordBearer*

DELETE /api/v1/accounts

User Login / Token Generation

Authenticates a user using their email (username) and password. Upon success, returns a **JWT Bearer Token** which must be included in the header of all protected requests.

REQUEST BODY SCHEMA: application/x-www-form-urlencoded
required

grant_type >	Grant Type (string) or Grant Type (null) (Grant Type)
username required	string (Username)
password required	string <password> (Password)
scope	string (Scope) Default: `" "`
client_id >	Client Id (string) or Client Id (null) (Client Id)
client_secret >	Client Secret (string) or Client Secret (null) <password> (Client Secret)

POST /api/v1/accounts/login

Response samples

200 422

Content type
application/json

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```
{
    "access_token": "eyJhbGciOiJIUzI1NiIs]
    "token_type": "bearer"
}
```

Lap Summaries

Import Telemetry from OpenF1

Fetches lap data from OpenF1 and saves it. Providing Year/Location/Session name automatically resolves the session_key.

AUTHORIZATIONS: > *OAuth2PasswordBearer*

QUERY PARAMETERS

`driver_number` required	integer (Driver Number) Examples: `driver_number=1` `driver_number=44` `driver_number=16` FIA Driver Number
`year` >	Year (integer) or Year (null) (Year) Examples: `year=2023` `year=2024` The four-digit race year
`location` >	Location (string) or Location (null) (Location) Examples: `location=Belgium` `location=Italy` `location=Silverstone` Grand Prix location
`session_name` >	SessionName (string) or Session Name (null) (Session Name) Session type
`session_key` >	Session Key (integer) or Session Key (null) (Session Key) Examples: `session_key=9141` `session_key=9158` Direct OpenF1 Session Key

POST /api/v1/lap_summaries/import_f… ⌄

Response samples

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Content type
application/json

Copy


```
{
    "imported": 44,
    "session_key": 9141
}
```


List Saved Lap Summaries

AUTHORIZATIONS: > *OAuth2PasswordBearer*

QUERY PARAMETERS

session_key >	Session Key (integer) or Session Key (null) (Session Key) Examples: `session_key=9141`
year >	Year (integer) or Year (null) (Year) Examples: `year=2023`
location >	Location (string) or Location (null) (Location) Examples: `location=Belgium` `location=Italy`
session_name >	SessionName (string) or Session Name (null) (Session Name)
driver_number >	Driver Number (integer) or Driver Number (null) (Driver Number) Examples: `driver_number=1` `driver_number=44`

GET /api/v1/lap_summaries

Response samples

200 422

Content type
application/json

Copy Expand all Collapse all


```
[
 - {
      "session_key": 9141,
      "year": 2023,
      "location": "Belgium",
      "session_name": "Race",
      "driver_number": 1,
      "lap_number": 1,
      "lap_duration": 107.54,
      "duration_sector_1": 30.45,
      "duration_sector_2": 48.12,
      "duration_sector_3": 28.97,
      "is_pit_out_lap": false,
      "is_pit_in_lap": false,
      "max_speed_kph": 315.5,
      "avg_speed_kph": 234.2,
      "i1_speed": 280,
      "i2_speed": 305,
      "st_speed": 315,
      "label": "Fastest Lap"
   }
]
```


Update Existing Summaries

AUTHORIZATIONS: > *OAuth2PasswordBearer*

QUERY PARAMETERS

`driver_number` required	integer (Driver Number) Examples: `driver_number=44`
`session_key` >	Session Key (integer) or Session Key (null) (Session Key) Examples: `session_key=9141`
`year` >	Year (integer) or Year (null) (Year) Examples: `year=2023`
`location` >	Location (string) or Location (null) (Location) Examples: `location=Belgium`
`session_name` >	SessionName (string) or Session Name (null) (Session Name)

PUT /api/v1/lap_summaries

Request samples

Payload

Content type
application/json

Copy


```
{
    "lap_duration": 106.9,
    "max_speed_kph": 320,
    "avg_speed_kph": 0,
    "i1_speed": 0,
    "i2_speed": 0,
    "st_speed": 0,
    "label": "Remodeled Data"
}
```


Response samples

200 422

Content type
application/json

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```
[
  - {
        "session_key": 9141,
        "year": 2023,
        "location": "Belgium",
        "session_name": "Race",
        "driver_number": 1,
        "lap_number": 1,
        "lap_duration": 107.54,
        "duration_sector_1": 30.45,
        "duration_sector_2": 48.12,
        "duration_sector_3": 28.97,
        "is_pit_out_lap": false,
        "is_pit_in_lap": false,
        "max_speed_kph": 315.5,
        "avg_speed_kph": 234.2,
        "i1_speed": 280,
        "i2_speed": 305,
        "st_speed": 315,
        "label": "Fastest Lap"
    }
]
```

Delete Summaries

AUTHORIZATIONS: > *OAuth2PasswordBearer*

QUERY PARAMETERS

`driver_number` required	integer (Driver Number) Examples: `driver_number=44`
`session_key` >	Session Key (integer) or Session Key (null) (Session Key) Examples: `session_key=9141`
`year` >	Year (integer) or Year (null) (Year) Examples: `year=2023`
`location` >	Location (string) or Location (null) (Location) Examples: `location=Belgium`
`session_name` >	SessionName (string) or Session Name (null) (Session Name)

DELETE /api/v1/lap_summaries

Response samples

422

Content type
application/json

Copy Expand all Collapse all

```
{
  - "detail": [
      + { … }
    ]
}
```

Data Analytics

Get Driver Pace Trends

Returns a chronological list of lap times for one or more drivers to visualize pace evolution over a session.

AUTHORIZATIONS: > *OAuth2PasswordBearer*

QUERY PARAMETERS

`year` required	integer (Year) Examples: `year=2023` The race year
`location` required	string (Location) Examples: `location=Belgium` `location=Monza` The circuit location
`session_name` >	SessionName (string) or Session Name (null) (Session Name) Filter by session type
`driver_numbers` >	Array of Driver Numbers (integers) or Driver Numbers (null) (Driver Numbers) Examples: `driver_numbers=44` `driver_numbers=63` `driver_numbers=1` List of driver numbers

GET /api/v1/analytics/pace-trend

Response samples

200 422

Content type
application/json

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```
{
    "year": 2024,
    "location": "Belgium",
  - "trends": [
      + { … }
    ]
}
```

Theoretical Ideal Lap Analysis

Calculates the 'Ideal Lap' (sum of best sectors) vs 'Actual Best' to identify potential improvement.

AUTHORIZATIONS: > *OAuth2PasswordBearer*

QUERY PARAMETERS

`year` required	integer (Year) Examples: `year=2023`
`location` required	string (Location) Examples: `location=Belgium`
`session_name` >	SessionName (string) or Session Name (null) (Session Name)
`driver_numbers` >	Array of Driver Numbers (integers) or Driver Numbers (null) (Driver Numbers) Examples: `driver_numbers=4` `driver_numbers=81`

GET /api/v1/analytics/ideal-laps

Response samples

200 422

Content type
application/json

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```
{
    "year": 2024,
    "location": "Belgium",
  - "overall_ideal_laps": [
      + { … }
    ],
  - "driver_ideal_laps": [
      + { … }
    ]
}
```

True Average Pace

Calculates average pace per driver, excluding pit and out-laps for representative performance.

AUTHORIZATIONS: > *OAuth2PasswordBearer*

QUERY PARAMETERS

`year` required	integer (Year) Examples: `year=2023`
`location` required	string (Location) Examples: `location=Belgium`
`session_name` >	SessionName (string) or Session Name (null) (Session Name)
`driver_numbers` >	Array of Driver Numbers (integers) or Driver Numbers (null) (Driver Numbers) Examples: `driver_numbers=1` `driver_numbers=11`

GET /api/v1/analytics/true-pace

Response samples

200 422

Content type
application/json

Copy Expand all Collapse all


```
[
  - {
      "driver_number": 1,
      "session_name": "Race",
      "true_average_pace": 109.12,
      "valid_laps_counted": 44
    }
]
```